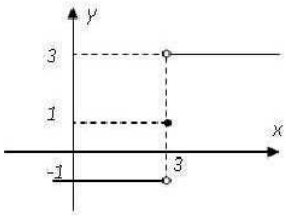
4.6 Lista de Exercícios

1. Sejam $f(x)$ as funções definidas pelos gráficos, encontre intuitivamente, se existir

a)



i. $\lim_{x\to 3^-} f(x)$

ii. $\lim_{x\to 3^+} f(x)$

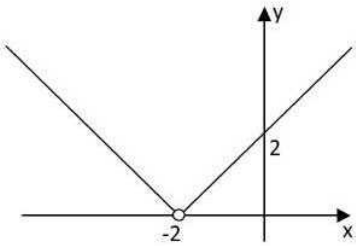
iii. $\lim_{x\to 3} f(x)$

iv. $\lim_{x\to -\infty} f(x)$

v. $\lim_{x\to +\infty} f(x)$

vi. $\lim_{x\to 4} f(x)$

b)



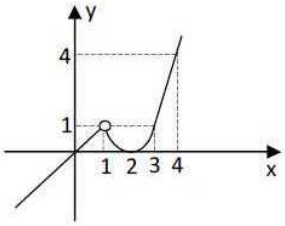
i. $\lim_{x\to -2^+} f(x)$

ii. $\lim_{x\to -2^-} f(x)$

iii. $\lim_{x\to -2} f(x)$

iv. $\lim_{x\to +\infty} f(x)$

c)

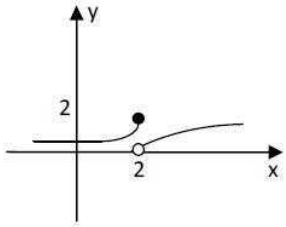


i. $\lim_{x\to 2^+} f(x)$

ii. $\lim_{x\to -\infty} f(x)$

iii. $\lim_{x\to 1} f(x)$

e)



i. $\lim_{x\to 2^+} f(x)$

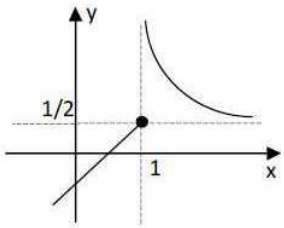
ii. $\lim_{x\to 2^-} f(x)$

iii. $\lim_{x\to 2} f(x)$

iv. $\lim_{x\to +\infty} f(x)$

v. $\lim_{x\to -\infty} f(x)$

d)



i. $\lim_{x\to 1^+} f(x)$

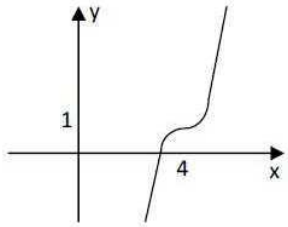
ii. $\lim_{x\to 1^-} f(x)$

iii. $\lim_{x\to 1} f(x)$

iv. $\lim_{x\to +\infty} f(x)$

v. $\lim_{x\to -\infty} f(x)$

f)



i. $\lim_{x\to 4^+} f(x)$

ii. $\lim_{x\to 4^-} f(x)$

iii. $\lim_{x\to 4} f(x)$

iv. $\lim_{x\to +\infty} f(x)$

v. $\lim_{x\to -\infty} f(x)$

2. Seja $f(x) = 2 + |5x - 1|$, calcule se existir os limites indicados e esboce o gráfico de $f(x)$:

   (a) $\lim\limits_{x \to 1/5^+} f(x)$ (b) $\lim\limits_{x \to 1/5^-} f(x)$ (c) $\lim\limits_{x \to 1/5} f(x)$

3. Seja $f(x) = \begin{cases} x^2 & , x \geq 0 \\ x - 2; & , x < 0. \end{cases}$

   Ache:

   (a) $\lim\limits_{x \to 0^-} f(x)$ (b) $\lim\limits_{x \to 0^+} f(x)$ (c) $\lim\limits_{x \to 0} f(x)$

4. Seja $f(x) = \dfrac{x^3 - 1}{x - 1}$

   (a) Ache $\lim\limits_{x \to 1} f(x)$

   (b) Esboce o gráfico de $y = f(x)$.

5. Seja $f(x) = \begin{cases} x - 1 & , x \leq 3 \\ 3x - 7 & , x > 3. \end{cases}$

   Esboçar o gráfico de $f(x)$ e calcular:

   (a) $\lim\limits_{x \to 3^-} f(x)$

   (b) $\lim\limits_{x \to 3^+} f(x)$

   (c) $\lim\limits_{x \to 3} f(x)$

   (d) $\lim\limits_{x \to 5^-} f(x)$

   (e) $\lim\limits_{x \to 5^+} f(x)$

   (f) $\lim\limits_{x \to 5} f(x)$

6. Seja $h(x) = \begin{cases} x^2 - 2x + 1 & , x \neq 3 \\ 7 & , x = 3. \end{cases}$

   Calcule $\lim\limits_{x \to 3} h(x)$.

7. Seja $f(x) = |x - 4|$. Esboçar o gráfico de $f(x)$ e calcular os limites indicados se existirem:

   (a) $\lim\limits_{x \to 4^+} f(x)$ (b) $\lim\limits_{x \to 4^-} f(x)$ (c) $\lim\limits_{x \to 4} f(x)$

8. Seja $f(x) = \dfrac{(x^2 - 25)}{x - 5}$. Calcule se existir $\lim\limits_{x \to 5} f(x)$.

9. Verifique se $\lim\limits_{x \to 1} \dfrac{1}{x - 1}$ existe.

10. Calcule os limites indeterminados:

    (a) $\lim\limits_{x \to -1} \dfrac{x^3 + 1}{x^2 - 1}$

    (b) $\lim\limits_{t \to -2} \dfrac{t^3 + 4t^2 + 4t}{(t + 2)(t - 3)}$

    (c) $\lim\limits_{x \to 5/2} \dfrac{2x^2 - 3x - 5}{2x - 5}$

    (d) $\lim\limits_{x \to -1} \dfrac{x^2 + 6x + 5}{x^2 - 3x - 4}$

    (e) $\lim\limits_{x \to 2} \dfrac{x^2 - 5x + 6}{x^2 - 12x + 20}$

    (f) $\lim\limits_{h \to 0} \dfrac{(4 + h)^2 - 16}{h}$

    (g) $\lim\limits_{h \to 0} \dfrac{\sqrt{25 + 3h} - 5}{h}$

    (h) $\lim\limits_{x \to 1} \dfrac{\sqrt{x} - 1}{\sqrt[3]{x} - 1}$

(i) $\lim\limits_{t\to 0} \dfrac{\sqrt[3]{27+t}-3}{t}$

(j) $\lim\limits_{x\to 2} \dfrac{\sqrt{x}-\sqrt{2}}{x-2}$

(k) $\lim\limits_{x\to a} \dfrac{\sqrt[3]{x}-\sqrt[3]{a}}{x-a}$

(l) $\lim\limits_{t\to 1} \dfrac{t^4+t^3+t-3}{t^2-1}$

(m) $\lim\limits_{h\to 2} \dfrac{\sqrt{h-1}-1}{h-2}$

(n) $\lim\limits_{h\to 1} \dfrac{\sqrt{h}-1}{h-1}$

(o) $\lim\limits_{x\to 2} \dfrac{x^3-5x^2+8x-4}{x^4-5x-6}$

(p) $\lim\limits_{t\to -1} \dfrac{\sqrt{t+5}-2}{t+1}$

(q) $\lim\limits_{t\to -1} \dfrac{t^2+5t+4}{t^2+3t+2}$

(r) $\lim\limits_{x\to 2} \dfrac{x^2+3x-10}{x^2-4}$

(s) $\lim\limits_{x\to 0} \dfrac{x^3+x}{x(x+1)}$

(t) $\lim\limits_{x\to 4} \dfrac{3x^2-17x+20}{x^2-2x-8}$

(u) $\lim\limits_{h\to 8} \dfrac{\sqrt[3]{h}-2}{8-h}$

(v) $\lim\limits_{x\to 1} \dfrac{\sqrt[3]{x}-1}{\sqrt[4]{x}-1}$

(w) $\lim\limits_{t\to 0} \dfrac{\sqrt{9-2t}-3}{t}$

(x) $\lim\limits_{x\to 2} \dfrac{\sqrt{x}-\sqrt{2}}{x-2}$

(y) $\lim\limits_{x\to 0} \dfrac{\sqrt{1+x}-\sqrt{1-x}}{x}$

(z) $\lim\limits_{x\to 2} \dfrac{x^2-4}{x-2}$

(aa) $\lim\limits_{x\to -1} \dfrac{x^2-1}{x^2+3x+2}$

(ab) $\lim\limits_{x\to -1} \dfrac{x^2+6x+5}{x^2-3x-4}$

(ac) $\lim\limits_{x\to 1} \dfrac{x^2-1}{x-1}$

(ad) $\lim\limits_{t\to 2} \dfrac{t^2-5t+6}{t-2}$

(ae) $\lim\limits_{x\to a} \dfrac{x^2+(1-a)x-a}{x-a}$

(af) $\lim\limits_{x\to 4} \dfrac{3-\sqrt{5+x}}{1-\sqrt{5-x}}$

(ag) $\lim\limits_{h\to 0} \dfrac{\sqrt[3]{8+h}-2}{h}$

(ah) $\lim\limits_{x\to 6} \dfrac{x^2-6x}{x-6}$

(ai) $\lim\limits_{x\to 9} \dfrac{x-9}{\sqrt{x}-3}$

(aj) $\lim\limits_{x\to 7} \dfrac{2-\sqrt{x-3}}{x^2-49}$

(ak) $\lim\limits_{x\to 4} \dfrac{3x^2-17x+20}{4x^2-25x+36}$

(al) $\lim\limits_{x\to 1} \dfrac{\sqrt[3]{x^2}-2\sqrt[3]{x}+1}{(x-1)^2}$

(am) $\lim\limits_{x\to 9} \dfrac{x-9}{\sqrt{x}-3}$

(an) $\lim\limits_{x\to 4} \dfrac{4-x}{2-\sqrt{x}}$

11. Calcular os limites no infinito:

(a) $\lim\limits_{x\to +\infty} (3x^2+4x^2-1)$

(b) $\lim\limits_{x\to +\infty} \dfrac{x+1}{x^2+1}$

(c) $\lim\limits_{t\to -\infty} \dfrac{-5t^3+2}{7t^3+3}$

(d) $\lim\limits_{x\to +\infty} \dfrac{x^2-1}{x-4}$

(e) $\lim\limits_{t\to \infty} \dfrac{t^2-2t+3}{2t^2+5t-3}$

(f) $\lim\limits_{x\to -\infty} \dfrac{5x^4-2x+1}{4x^4+3x+2}$

(g) $\lim\limits_{x\to -\infty} \dfrac{x^2-1}{x^3-4}$

(h) $\lim\limits_{t\to \infty} \dfrac{t^5+t^3}{t^2-t^6}$

(i) $\lim\limits_{x\to -\infty} \dfrac{2x^3+3x-1}{3x^3-2}$

(j) $\lim\limits_{x\to +\infty} \dfrac{1}{x-12}$

(k) $\lim\limits_{x\to+\infty}(3x^3+4x^2-1)$

(l) $\lim\limits_{x\to+\infty}\left(2-\dfrac{1}{x}+\dfrac{4}{x^2}\right)$

(m) $\lim\limits_{t\to+\infty}\dfrac{t+1}{t^2+1}$

(n) $\lim\limits_{x\to-\infty}\dfrac{x-2}{x^2+2x+1}$

(o) $\lim\limits_{t\to+\infty}\dfrac{t^2-2t+3}{2t^2+5t-3}$

(p) $\lim\limits_{x\to+\infty}\dfrac{2x^5-3x^3+2}{-x^2+7}$

(q) $\lim\limits_{x\to-\infty}\dfrac{3x^5-x^2+7}{2-x^2}$

(r) $\lim\limits_{x\to+\infty}\dfrac{x^2+3x+1}{x}$

(s) $\lim\limits_{x\to-\infty}\dfrac{\sqrt{5x^2-2}}{x+3}$

(t) $\lim\limits_{x\to+\infty}\dfrac{\sqrt{x^2+1}}{x+1}$

(u) $\lim\limits_{x\to-\infty}\dfrac{x^3-2x+1}{x^2-1}$

(v) $\lim\limits_{s\to+\infty}\dfrac{8-s}{\sqrt{s^2+7}}$

(w) $\lim\limits_{x\to+\infty}\dfrac{10x^2-3x+4}{3x^2-1}$

(x) $\lim\limits_{x\to+\infty}\dfrac{\sqrt{2x^2-7}}{x+3}$

12. Calcular os limites infinitos

(a) $\lim\limits_{x\to3^+}\dfrac{x}{x-3}$

(b) $\lim\limits_{x\to6^-}\dfrac{x+6}{x^2-36}$

(c) $\lim\limits_{x\to4^+}\dfrac{3-x}{x^2-2x-8}$

(d) $\lim\limits_{x\to2^-}\dfrac{x}{x^2-4}$

(e) $\lim\limits_{x\to5^+}\dfrac{x+5}{25-x^2}$

(f) $\lim\limits_{x\to3^-}\dfrac{x}{9-x^2}$

(g) $\lim\limits_{y\to6^+}\dfrac{y+6}{y^2-36}$

(h) $\lim\limits_{x\to4^+}\dfrac{3-x}{x^2-2x-8}$

(i) $\lim\limits_{x\to3^-}\dfrac{1}{|x-3|}$

(j) $\lim\limits_{x\to3^+}\dfrac{1}{|x-3|}$

13. Seja $f(x)=\begin{cases} x^2, & x<0,\\ 3, & x=0,\\ \dfrac{1}{x}, & x>0.\end{cases}$ Esboce o gráfico, e calcule se existirem:

a) $\lim\limits_{x\to+\infty}f(x)$

b) $\lim\limits_{x\to-\infty}f(x)$

c) $\lim\limits_{x\to0^+}f(x)$

d) $\lim\limits_{x\to0^-}f(x)$

e) $\lim\limits_{x\to0}f(x)$

14. Calcule os seguintes limites aplicando os limites fundamentais

(a) $\lim\limits_{x\to0}\dfrac{\operatorname{sen}9x}{x}$

(b) $\lim\limits_{x\to0}\dfrac{\operatorname{sen}4x}{3x}$

(c) $\lim\limits_{x\to0}\dfrac{\operatorname{sen}5x}{\operatorname{sen}2x}$

(d) $\lim\limits_{x\to0}\dfrac{\operatorname{sen}4x}{5x}$

(e) $\lim\limits_{x\to0}\dfrac{x}{\operatorname{sen}x}$

(f) $\lim\limits_{x\to0}\dfrac{\tan ax}{x}$

(g) $\lim\limits_{x\to0}\dfrac{1-\cos x}{x^2}$

(h) $\lim\limits_{x\to0}\dfrac{x-\operatorname{sen}x}{\operatorname{sen}x}$

(i) $\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{3n}$

(j) $\lim_{x\to\infty}\left(1+\frac{2}{x}\right)^{x}$

(k) $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x+5}$

(l) $\lim_{x\to\infty}\left(\frac{x}{1+x}\right)^{x}$

(m) $\lim_{x\to\infty}\left(1-\frac{6}{x}\right)^{x}$

(n) $\lim_{x\to 2}\frac{5^{x-2}-1}{x-2}$

(o) $\lim_{x\to -3}\frac{4^{\frac{x+3}{5}}-1}{x+3}$

(p) $\lim_{x\to 2}\frac{5^{x}-25}{x-2}$

15. Investigue a continuidade nos pontos indicados:

(a) $f(x)=\begin{cases} 2x-3 \text{ se } x\leq 2 \\ x^2 \quad \text{se } x>2. \end{cases}$ em $x=2$

(b) $f(x)=\begin{cases} 3x^2+5 \text{ se } x\neq 1 \\ 6 \quad \text{se } x=1. \end{cases}$ em $x=1$

(c) $f(x)=\begin{cases} x^2 \quad \text{se } x\geq 1 \\ 1-|x| \quad \text{se } x<1. \end{cases}$ em $x=1$

(d) $f(x)=\begin{cases} \frac{x^2-4}{x-2} \quad \text{se } x\neq 2 \\ 0 \quad \text{se } x=2. \end{cases}$ em $x=2$

(e) $f(x)=\begin{cases} 1-x^2 \quad \text{se } x<1 \\ 1-|x| \quad \text{se } x>1 \\ 1 \quad \text{se } x=1. \end{cases}$ em $x=1$

(f) $f(x)=\begin{cases} \frac{\operatorname{sen} x}{x} \quad \text{se } x\neq 0 \\ 0 \quad \text{se } x=0. \end{cases}$ em $x=0$

(g) $f(x)=\begin{cases} \frac{x^3-8}{x^2-4} \quad \text{se } x\neq 2 \\ 3 \quad \text{se } x=2. \end{cases}$ em $x=2$

(h) $f(x)=\frac{x^2-3x+7}{x^2+1}$, em $x=2$

(i) $f(x)=\frac{2}{3x^2+x^3-x-3}$, em $x=-3$

16. Determinar, se existirem, os valores de $x\in D(f)$, nos quais a função $f(x)$ não é contínua:

(a) $f(x)=\begin{cases} \frac{x}{x^2-1} \quad , x^2\neq 1 \\ 0 \quad , x=-1. \end{cases}$

(b) $f(x)=\begin{cases} \sqrt{x^2+5x+6} \quad , x<-3 \text{e} x>-2 \\ -1 \quad , -3\leq x\leq -2. \end{cases}$

(c) $f(x)=\begin{cases} \frac{3}{x-1} \quad , x\neq 1 \\ 3 \quad , x=-1. \end{cases}$

(d) $f(x)=\begin{cases} 1-\cos x \quad , x<0 \\ x^2+1 \quad , x\geq 0. \end{cases}$

(e) $f(x) = \begin{cases} \dfrac{x^2 - 3x + 4}{x - 1} & , x \neq 1 \\ 1 & , x = 1. \end{cases}$

(f) $f(x) = \dfrac{x}{x^2 - 1}$

(g) $f(x) = \dfrac{x}{(x-3)(x+7)}$

(h) $f(x) = \sqrt{(3-x)(6-x)}$

(i) $f(x) = \dfrac{x^2 + 3x - 1}{x^2 - 6x + 10}$

17. Calcule p de modo que as funções abaixo sejam contínuas.

(a) $f(x) = \begin{cases} x^2 + px + 2 & , x \neq 3 \\ 3 & , x = 3. \end{cases}$

(b) $f(x) = \begin{cases} x + 2p & , x \leq -1 \\ p^2 & , x > -1. \end{cases}$

Gabarito

1. (a) (i) -1
   (ii) 3
   (iii) Não existe.
   (iv) -1
   (v) 3
   (vi) 3
   (b) (i) 0
   (ii) 0
   (iii) 0
   (iv) $+\infty$
   (c) (i) 0
   (ii) $-\infty$
   (iii) 1
   (d) (i) $+\infty$
   (ii) $\dfrac{1}{2}$
   (iii) Não existe.
   (iv) $\dfrac{1}{2}$
   (v) $-\infty$
   (e) (i) 0
   (ii) 2
   (iii) Não existe.
   (iv) 2
   (v) 0
   (f) (i) 1
   (ii) 1
   (iii) 1
   (iv) $+\infty$
   (v) $-\infty$

2. (a) 2
   (b) 2
   (c) 2

3. (a) -2
   (b) 0
   (c) Não existe.

4. (a) 3

5. (a) 2
   (b) 2
   (c) 2
   (d) 8
   (e) 8
   (f) 8

6. 4

7. (a) 0
   (b) 0
   (c) 0

8. 10

9. O limite não existe.

10. (a) $-\frac{3}{2}$
    (b) 0
    (c) $\frac{7}{2}$
    (d) $-\frac{4}{5}$
    (e) $\frac{1}{8}$
    (f) 8
    (g) $\frac{3}{10}$
    (h) $\frac{3}{2}$
    (i) $\frac{1}{27}$
    (j) $\frac{\sqrt{2}}{4}$
    (k) $\frac{1}{3\sqrt[3]{a^2}}$
    (l) 4
    (m) $\frac{1}{2}$
    (n) 1
    (o) 0
    (p) $\frac{1}{4}$
    (q) 3
    (r) $\frac{7}{4}$
    (s) 1
    (t) $\frac{7}{6}$
    (u) $-\frac{1}{12}$
    (v) $\frac{4}{3}$
    (w) $-\frac{1}{3}$
    (x) $\frac{\sqrt{2}}{4}$
    (y) 1
    (z) 4
    (aa) -2
    (ab) $-\frac{4}{5}$
    (ac) 2
    (ad) -1
    (ae) $a+1$
    (af) $-\frac{1}{3}$
    (ag) $\frac{1}{12}$
    (ah) 6
    (ai) 6
    (aj) $-\frac{1}{56}$
    (ak) 1
    (al) $\frac{1}{9}$
    (am) 6
    (an) 4

11. (a) $+\infty$
    (b) 0
    (c) $-\frac{5}{7}$
    (d) $+\infty$
    (e) $\frac{1}{2}$
    (f) $\frac{5}{4}$
    (g) 0
    (h) 0
    (i) $\frac{2}{3}$
    (j) 0
    (k) $+\infty$
    (l) 2
    (m) 0
    (n) 0
    (o) $\frac{1}{2}$
    (p) $-\infty$
    (q) $+\infty$
    (r) $+\infty$
    (s) $\sqrt{5}$
    (t) 1
    (u) $-\infty$
    (v) -1
    (w) $\frac{10}{3}$
    (x) $\sqrt{2}$

12. (a) $+\infty$
    (b) $-\infty$
    (c) $-\infty$
    (d) $-\infty$
    (e) $-\infty$
    (f) $+\infty$
    (g) $+\infty$
    (h) $-\infty$
    (i) $+\infty$
    (j) $+\infty$

13. (a) 0
    (b) $+\infty$
    (c) $+\infty$
    (d) 0
    (e) Não existe.

14. (a) 9
    (b) $\frac{4}{3}$
    (c) $\frac{5}{2}$
    (d) $\frac{4}{5}$
    (e) 1
    (f) a
    (g) $\frac{1}{2}$
    (h) 0
    (i) e^3
    (j) e^2
    (k) e
    (l) $1/e$
    (m) $1/e^6$
    (n) $\ln 5$
    (o) $\frac{2}{5}\ln 2$
    (p) $25\ln 5$

15. (a) Não contínuo.
    (b) Não contínuo.
    (c) Não contínuo.
    (d) Não contínuo.
    (e) Não contínuo.
    (f) Não contínuo.
    (g) Contínuo.
    (h) Contínuo.
    (i) Não contínuo.

16. (a) -1
    (b) -3 e -2
    (c) 1
    (d) 0
    (e) 1
    (f) -1 e 1
    (g) -7 e 3
    (h) 3 e 6
    (i) Não existe.

17. (a) $\dfrac{-8}{3}$
    (b) 1